Adversarial Robustness in Fake Quantum Simulators

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Abstract

This paper investigates the performance scalability and adversarial robustness of Quantum Machine Learning (QML) models deployed on noise-model-based fake simulators. We conduct a dual-phased study, first benchmarking the computational throughput of Qiskit's Aer simulation engine across varying hardware architectures, and second, evaluating the effectiveness of Projected Gradient Descent (PGD) attacks and adversarial retraining strategies under realistic noise conditions. Our results quantify the runtime tradeoffs for medium-scale (8-qubit) simulations and demonstrate that high adversarial-to-benign retraining ratios (50/50) are essential for achieving practical model robustness in noisy environments.

1 Introduction

The field of Quantum Machine Learning (QML) has seen rapid growth, but the susceptibility of Quantum Neural Networks (QNNs) to adversarial perturbations remains a critical challenge. As established by Lu et al. [LDD20], quantum classifiers are vulnerable to gradient-based attacks, mirroring the vulnerabilities found in classical deep learning. Among these, the Projected Gradient Descent (PGD) algorithm has emerged as a particularly potent white-box attack capable of circumventing simple defenses [Web+25].

While much research has focused on ideal simulators, evaluating these attacks on the Noisy Intermediate-Scale Quantum (NISQ) devices used today is essential. Qiskit's fake backends offer a vital tool for this purpose, providing noise models derived from real system snapshots [Ren+24; Qis24]. However, the fidelity of these models and the generalization properties of adversarially trained quantum models under specific noise profiles require further investigation [DuY+24]. This project builds upon the foundational work on QML robustness by Maußner and Reers [MR25], extending their analysis to a broader range of simulated architectures.

The software implementation for this study, including all benchmarking scripts and the PennyLane-Qiskit integration, is available in the project repository [Mau26].

In this paper, we aim to:

1. Evaluate the performance and scalability of Qiskit's Fake Backends (V2) for density matrix simulations, focusing on thread parallelization and Aer method selection.

2. Quantify the runtime scaling behavior from 4 to 8 qubits to establish the feasibility of medium-scale noisy simulation.

3. Benchmark the effectiveness of the PGD attack algorithm across multiple 4-qubit hardware-mimicking architectures (`FakeGuadalupeV2`, `FakeLimaV2`, `FakeJakartaV2`).

4. Analyze the impact of varying adversarial retraining ratios on the robustness of QNNs under realistic noise conditions.

2 Methodology

In this work, we evaluate the adversarial robustness of Quantum Machine Learning (QML) models using noisy simulators that mimic real quantum hardware. Our methodology leverages the Qiskit [Qis23] ecosystem and the PennyLane [Ber+18] framework. The study is divided into two primary phases: performance benchmarking and robustness evaluation.

2.1 Performance Benchmarking of Noisy Simulators

To optimize simulation throughput for noise-aware training, we investigate the following performance factors:

- **Parallelization Strategies:** We evaluate the impact of `max_parallel_threads` and `max_parallel_experiments` on simulation speed. High-performance multicore configurations are tested to mitigate the 4^n state-space complexity of density matrix simulations. This includes a comprehensive profiling of PennyLane-integrated circuits across varying thread counts to identify optimal resource allocation strategies for gradient-heavy workloads.
- **Aer Simulation Methods:** We benchmark three distinct simulation methods in Qiskit Aer: `automatic`, `matrix_product_state` (MPS), and `statevector`. This helps identify the most efficient solver for CNOT-heavy circuits under noise.
- **Runtime Scaling Estimation:** We develop a quantitative scaling model to predict the runtime of medium-scale experiments (8 qubits) based on measured 4-qubit benchmarks, accounting for parameter count, circuit depth, and parallelization speedup.

2.2 Noise-Aware Adversarial Robustness Evaluation

The evaluation of noise-aware adversarial robustness follows a multi-stage experimental workflow, as illustrated in Figure 1. The process begins with the acquisition and reduction of the `plus-minus` dataset to ensure computational feasibility. Simultaneously, a performance benchmarking phase identifies the optimal backend configurations. These inputs feed into the QNN construction phase, followed by supervised training on benign data. Once the model converges, we generate adversarial examples using PGD to evaluate the initial robustness. Finally, we implement defense strategies like adversarial retraining and evaluate the resulting robust model using holistic metrics.

The second phase evaluates the security of QNNs under the following experimental framework:

- **QNN Architecture:** We utilize a Strongly Entangling Layer ansatz with data reuploading (3×) across 4, 6, and 8-qubit configurations. The model is integrated into a PennyLane-PyTorch hybrid pipeline.
- **Dataset Reduction:** To ensure feasibility in noisy environments, we use a reduced subset of the `plus-minus` dataset (200 training and 50 test samples).
- **Adversarial Attacks:** We generate white-box perturbations using the Projected Gradient Descent (PGD) algorithm ($\epsilon = 0.1, \alpha = 0.01, 10$ iterations).

- **Defense Strategies:** We investigate adversarial retraining with varying benign-to-adversarial sample ratios. Specifically, we compare a *sparse* 10% baseline (where only 10% of the training samples are adversarial) with a *balanced* 50/50 split. In the balanced approach, the training dataset is effectively doubled by adding an equal number of PGD-generated adversarial examples to the original benign dataset, ensuring the QNN learns to minimize the adversarial loss as a primary objective.

- **Structural Variations:** We test several architectural and optimization strategies to enhance resilience:

  - **Non-Linear Feature Mapping:** We apply a `tanh` activation function to the input features prior to angle encoding via `StronglyEntanglingLayers`. This introduces higher-order non-linearity into the classical-quantum interface to assess its impact on adversarial susceptibility.
  - **Lipschitz Gradient Regularization:** We incorporate a gradient-magnitude penalty into the loss function, where the penalty is proportional to the norm of the gradient of the QNN output with respect to the input features. This Lipschitz-like constraint explicitly bounds the model's sensitivity to small input perturbations.
  - **Learning Rate Scheduling:** We use a `StepLR` scheduler to stabilize the noisy loss landscape, alternating between high learning rates for exploration and periodic decays for fine-tuning.

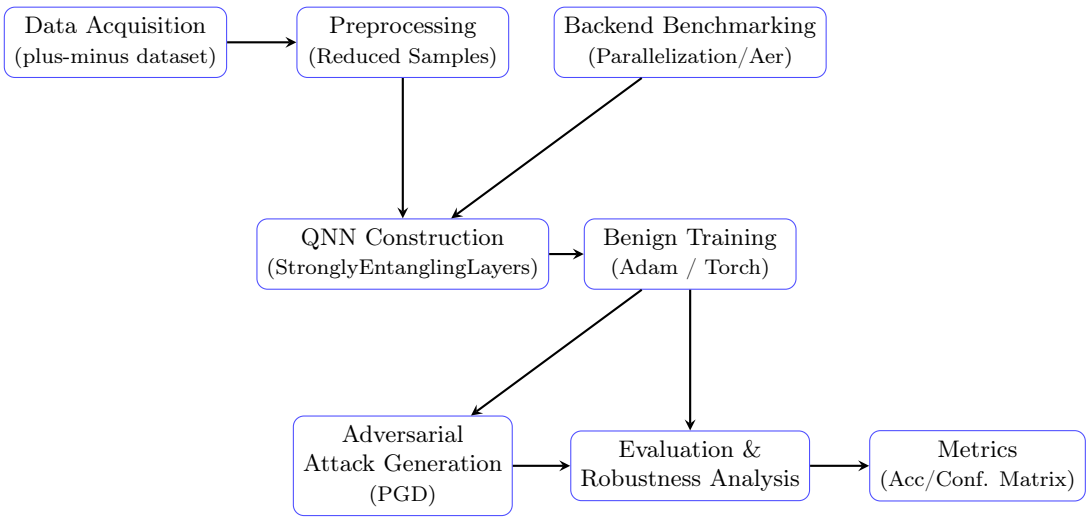

Figure 1: Experimental workflow for noise-aware adversarial evaluation. The pipeline consists of: (1) Acquisition and reduction of the `plus-minus` dataset, (2) Backend benchmarking to optimize simulation threads and methods, (3) QNN construction using Strongly Entangling Layers, (4) Model training on benign data, (5) Adversarial example generation via Projected Gradient Descent (PGD), and (6) Robustness analysis using accuracy and confusion matrices.

3 Results

Our experimental evaluation reveals significant performance tradeoffs and robustness variations when executing adversarial attacks across different simulated noisy environments.

3.1 Performance Benchmarking Results

The benchmarking of the `Qiskit Aer` engine on the `FakeGuadalupeV2` backend, illustrated in Figure 2, demonstrates that the `automatic` method consistently identifies the most efficient execution path. In contrast, the `matrix_product_state` (MPS) method suffers from high computational overhead in CNOT-heavy circuits, even at low qubit counts ($n \leq 8$).

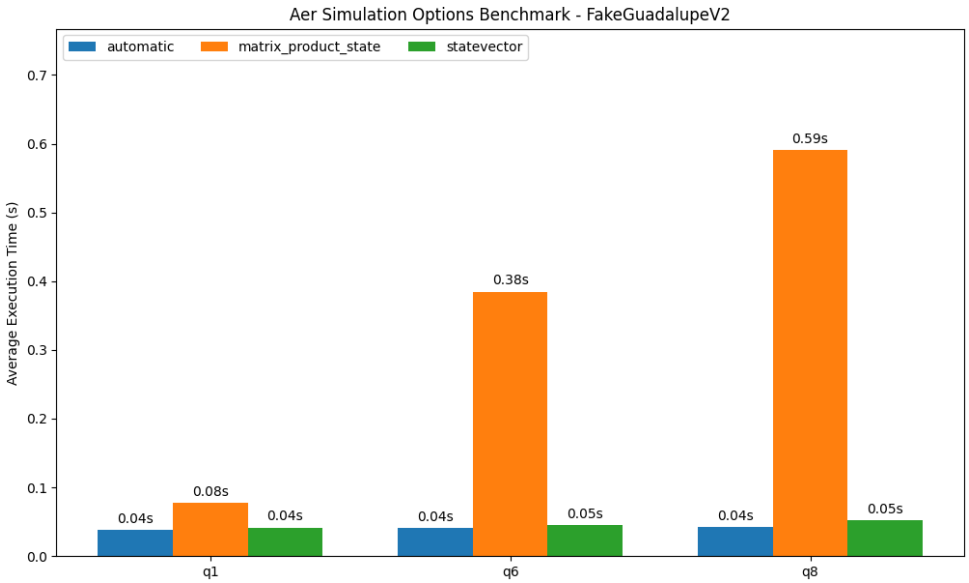

Figure 2: Benchmarking of simulation methods in Qiskit Aer on the `FakeGuadalupeV2` backend. The `automatic` method (blue) identifies the optimal solver for specific circuit configurations, significantly outperforming the `matrix_product_state` (MPS) approach as the qubit count scales from 4 to 8 qubits.

The broader impact of multicore execution on the PennyLane hybrid interface is captured in Figure 3. This comprehensive profiling compares execution runtimes across multiple noisy backends, illustrating the overhead introduced by the density matrix evolution.

The results in Figure 3 demonstrate that while multithreading provides significant speedups for individual circuit executions, the PennyLane-Qiskit communication layer introduces a non-negligible serial bottleneck. For smaller devices like `FakeLimaV2`, the simulation is relatively efficient even with minimal parallelization. However, for higher-qubit backends such as `FakeGuadalupeV2`, the optimal configuration requires balancing `max_parallel_threads` (for C++-level acceleration) and `max_parallel_experiments` (for batching gradient circuits).

Detailed parallelization profiling shows that increasing `max_parallel_threads` yields an average speedup of $1.9\times$ for 8-qubit circuits. However, runtime estimation models reveal a steep scaling factor ($10.86\times$) when doubling the qubit count from 4 to 8, primarily driven by the exponential growth in density matrix complexity and the linear increase in parameters and circuit depth.

3.2 Noise-Aware Adversarial Robustness Results

The adversarial evaluation was first conducted on several 4-qubit backend architectures. Table 1 summarizes the classification performance under benign conditions, PGD attack, and standard adversarial retraining (10% ratio).

A critical finding was the impact of the adversarial-to-benign retraining ratio. While the 10% ratio provided only minor robustness recovery (44% on Guadalupe), increasing

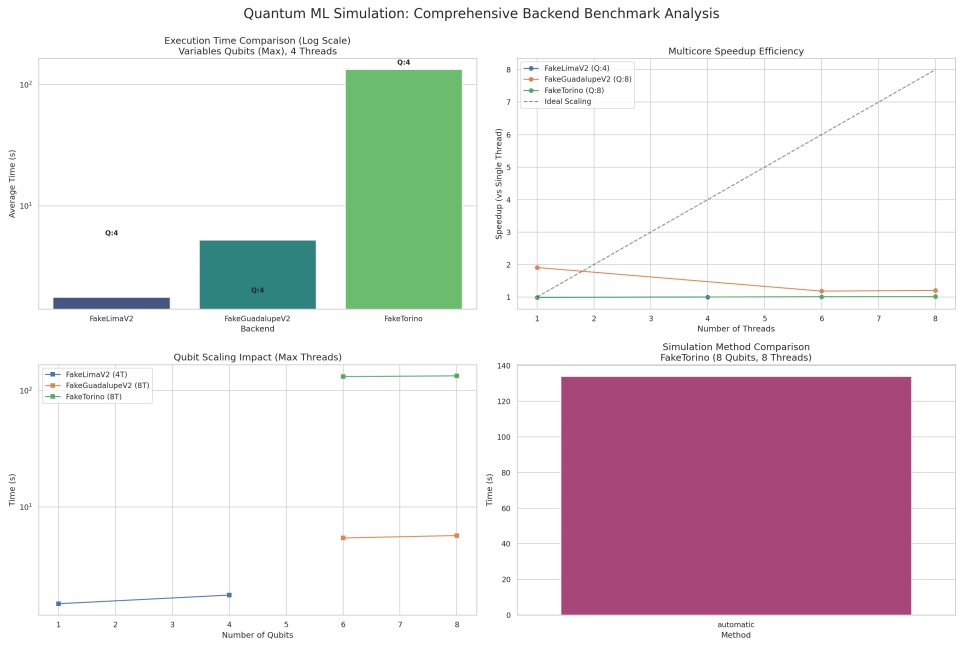

Figure 3: Comprehensive comparison of simulation runtimes via the PennyLane interface across multiple noisy backends (`FakeGuadalupeV2`, `FakeLimaV2`, `FakeJakartaV2`). The plots illustrate the impact of parallelization configurations (`max_parallel_threads`) and highlight the communication overhead introduced by the PennyLane-Qiskit bridge.

Backend (4-Qubit)	**Benign Acc**	**Adversarial Acc**	**Robust Acc (10%)**
FakeGuadalupeV2	0.96	0.40	0.44
FakeLimaV2	0.92	0.26	0.52
FakeJakartaV2	0.96	0.12	0.48

Table 1: Performance metrics of a 4-qubit QNN across different simulated noisy environments. Benign accuracy (clean data) is compared against adversarial accuracy under a PGD attack ($\epsilon = 0.1$) and robust accuracy after standard adversarial retraining with a 10% ratio of adversarial samples.

the ratio to a 50/50 split dramatically improved the robust accuracy to **84.00%**, as shown in Figure 4.

Beyond retraining ratios, we evaluated several architectural and optimization strategies to enhance model resilience. Table 2 provides a comprehensive comparison of these methods on the `FakeGuadalupeV2` backend.

The results in Table 2 highlight several key insights. First, the introduction of non-linear feature mappings (`tanh`) increases the gradient's susceptibility to perturbations, leading to a sharp decline in adversarial accuracy (12%) compared to the linear baseline. Second, the `StepLR` scheduler remains a foundational benchmark (*TBD*), while Lipschitz regularization demonstrates a significant improvement in robust accuracy (64%) compared to the golden sample baseline (44%), successfully stabilizing the loss landscape during noisy gradient descent. Most significantly, the transition from a sparse adversarial retraining ratio (10%) to a balanced 50/50 split emerges as the most effective defense mechanism, yielding a 40% absolute improvement in robust accuracy. This suggests that in hardware-mimicking environments, the diversity and volume of adversarial examples during the fine-tuning phase are more critical for security than architectural complexity alone.

Strategy	Benign Acc	Adversarial Acc	Robust Acc
Golden Sample (10% Ratio)	0.96	0.40	0.44
Non-Linear Map (`tanh`)	0.96	0.12	$0.44^{\dagger}$
Retraining Ratio (50/50)	0.96	0.14	**0.84**
LR Scheduler (`StepLR`)	0.96	0.14	0.12
Lipschitz Regularization	0.94	0.44	0.64

Table 2: Comparative analysis of noise-aware robustness strategies on the `FakeGuadalupeV2` backend. Strategies include non-linear feature mappings (`tanh`), learning rate scheduling (`StepLR`), and varying adversarial retraining ratios. The 50/50 retraining ratio achieves the highest robust accuracy (84%), while † denotes results requiring further epoch-wise verification.

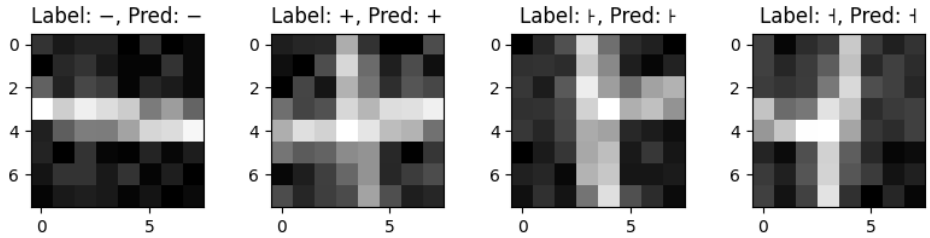

Figure 4: Comparison of model performance on `FakeGuadalupeV2` after balanced (50/50) adversarial retraining. This strategy results in a substantial recovery of robustness, increasing robust accuracy from 44% (sparse 10% ratio) to 84%, demonstrating that dataset balance is critical for security in hardware-mimicking environments.

Finally, we observed that non-linear feature mappings (`tanh`) do not inherently confer robustness; in fact, they exhibited higher vulnerability, with adversarial accuracy dropping to 12%. Lipschitz regularization and learning rate scheduling showed potential for stabilizing noisy loss landscapes, with Lipschitz regularization providing a 20% absolute improvement in robustness over the standard retraining baseline. However, full convergence remains computationally intensive at larger scales.

4 Conclusion

This work evaluated the performance scalability and adversarial robustness of Quantum Machine Learning models across several simulated noisy environments. Our results demonstrate that while density matrix simulation of noisy 8-qubit devices is computationally intensive, the implementation of multicore parallelization and the selection of optimized Aer solver methods make such studies feasible on modern classical hardware. We establish a quantitative scaling factor of $10.86\times$ for runtime increase when doubling the qubit count from 4 to 8.

Regarding adversarial security, we confirm that white-box perturbations generated via PGD can reduce classification accuracy from near-perfect levels to as low as 12% in noisy environments. Our investigation into defense strategies reveals that standard adversarial retraining provides only modest recovery when using low ratios of adversarial samples. In contrast, a 50/50 adversarial-to-benign retraining split was shown to be highly effective, restoring robust accuracy to 84%. We also demonstrate that Lipschitz gradient regularization provides a substantial 20% absolute improvement in robustness over the standard retraining baseline. Furthermore, we find that architectural choices such as non-linear feature mappings can inadvertently increase vulnerability to gradient-based attacks. These

findings emphasize the necessity of co-designing QNN architectures and training protocols to achieve practical security in the NISQ era.

Outlook Future research will explore the transferability of adversarial robustness across diverse quantum architectures. Specifically, we aim to investigate "cross-backend" resilience, where QNN models are trained on one QPU simulator and subsequently evaluated on different hardware-mimicking environments to assess the impact of varying noise profiles. Additionally, we plan to integrate advanced model optimization techniques, such as Approximate State Preparation (ASP) [Wes+23], to further enhance the defensive capabilities of QNNs against sophisticated adversarial perturbations.

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